Reductions for the finite implication E677 $\models_{\text{fin}}$ E255

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May 2026

Abstract

The Equational Theories Project leaves open, up to duality, the finite implication E677 $\models_{\text{fin}}$ E255 for magmas, while the corresponding unrestricted implication is known to be false. This note records a checked package of reductions for that remaining finite problem. The main output is a route graph: the target law at a point is equivalent to a unique right fixed point, to a left-division identity called Lemma L, and to several constrained collision and counterexample conditions. The note also isolates false trails that should not be reused, including a previously suspected quotient family and a full-shift law. No proof or finite counterexample is claimed.

2020 Mathematics Subject Classification. Primary 08A05; Secondary 08B05, 03C13, 68V15.

Keywords. Finite magma, equational law, finite implication, left division, equational theories.

1 Introduction

A *magma* is a set M equipped with a binary operation $\diamond$. The two laws studied here are

$$\begin{aligned} \text{E677}: &\qquad x = y \diamond \big(x \diamond ((y \diamond x) \diamond y)\big), \\ \text{E255}: &\qquad x = ((x \diamond x) \diamond x) \diamond x. \end{aligned}$$

The problem is the finite implication

$$\text{E677} \models_{\text{fin}} \text{E255},$$

that is, whether every finite magma satisfying E677 also satisfies E255. The unrestricted implication E677 $\models$ E255 is false. The Equational Theories Project (ETP) paper records the finite problem as the only unresolved finite implication up to duality, equivalently together with the dual implication E2910 $\models_{\text{fin}}$ E47, and tentatively conjectures the implication to be false [1, Problem 8.1]. The ETP blueprint chapter on Equation 677 contains the local structure theory and the known no-go statements for standard counterexample families [4]; the dashboard records the graph status [3].

This note is not a computational search report. Its purpose is to make the remaining mathematical problem sharper. The reductions below are organized around a fixed element x of a finite E677-magma. They show that the target law at x is the same as a single right fixed point, that any such fixed-point witness is unique, and that the same target can be expressed as a left-division identity (Lemma L). The complementary sections describe what a collision proof or a finite counterexample would have to supply.

2 Preliminaries and external input

For $a \in M$, write

$$L_a(t) = a \diamond t, \qquad R_a(t) = t \diamond a, \qquad S(t) = t \diamond t.$$

If L_a is bijective, write $a \backslash b$ for the unique element u satisfying $a \diamond u = b$. All products in this note are non-associative and are parenthesized when there is any possible ambiguity.

The finite hypothesis first enters through the following standard ETP fact.

Proposition 1 (Left division in finite E677-magmas)**.** *Let M be a finite magma satisfying* E677. *Then every left translation L_a is bijective. Moreover*

$$a \backslash b = b \diamond ((a \diamond b) \diamond a)$$

for all $a, b \in M$.

Proof. Law E677 gives

$$b = a \diamond \big(b \diamond ((a \diamond b) \diamond a)\big).$$

Thus L_a is surjective. Since M is finite, L_a is bijective, and the displayed preimage formula is exactly the formula for $a \backslash b$. □

Lemma 2 (Division calculus)**.** *Let M be a finite magma satisfying* E677. *Then*

$$a \backslash (a \backslash b) = (a \backslash b) \diamond (b \diamond a), \qquad b \backslash (a \backslash b) = (a \diamond b) \diamond a$$

for all $a, b \in M$.

Proof. For the first identity, Proposition 1, applied to $a \backslash b$ in place of b, gives

$$a \backslash (a \backslash b) = (a \backslash b) \diamond ((a \diamond (a \backslash b)) \diamond a) = (a \backslash b) \diamond (b \diamond a),$$

because $a \diamond (a \backslash b) = b$. For the second identity, Proposition 1 gives

$$b \diamond ((a \diamond b) \diamond a) = a \backslash b,$$

so uniqueness of left division by b gives $b \backslash (a \backslash b) = (a \diamond b) \diamond a$. □

The ETP chapter also records several strategic facts that are used here only as external input. The unrestricted implication E677 $\not\models$ E255 is witnessed by the free E677-magma [4, Section 13.2]. Linear models satisfying E677 satisfy E255, and affine-fiber extensions over such models do not produce counterexamples [4, Lemmas 13.3–13.4]. The same chapter gives finite E677-magmas that are not right-cancellative [4, Section 13.1]. Hence no argument below may use right cancellation, associativity, an identity element, or group or quasigroup intuition unless such structure has first been derived. The finite-map lemmas in the ETP finite chapter are a natural source of future tools [5, Lemmas 5.5–5.7].

Lemma 3 (Transformed identity)**.** *Let M be a finite magma satisfying* E677. *Then*

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y)$$

for all $y, z \in M$.

Proof. Substitute $y \diamond z$ for x in E677. This gives

$$y \diamond z = y \diamond \big((y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y)\big).$$

Cancel the common left factor y using Proposition 1. □

Lemma 4 (Key identity)**.** *Let M be a finite magma satisfying* E677. *Then*

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z))$$

for all $y, z \in M$.

Proof. Lemma 3 gives

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y).$$

On the other hand, applying E677 with the parameter $y \diamond z$ gives

$$z = (y \diamond z) \diamond \big(z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z))\big).$$

Left cancellation by $y \diamond z$ gives the identity. □

3 The orbit package

Fix a finite E677-magma M and an element $x \in M$. Define

$$c_0 = x, \qquad c_{k+1} = x \diamond c_k,$$

and let d be the minimal period of this L_x-orbit, so that $c_d = x$ and $c_0, \ldots, c_{d-1}$ are pairwise distinct. Indices on the c_k are read modulo d. Put

$$d_k = c_k \diamond x.$$

Lemma 5 (Orbit recurrence)**.** *For every k,*

$$c_{k-1} = c_k \diamond d_{k+1}.$$

Consequently $d_1 = c_{d-2}$.

Proof. Apply E677 with $y = x$ and the law variable x replaced by c_k:

$$c_k = x \diamond \big(c_k \diamond ((x \diamond c_k) \diamond x)\big) = x \diamond (c_k \diamond d_{k+1}).$$

Since also $c_k = x \diamond c_{k-1}$, left cancellation gives the recurrence. Taking $k = 0$ gives $c_{d-1} = x \diamond d_1$, so $d_1 = x \backslash c_{d-1} = c_{d-2}$. □

Lemma 6 (Basic left-division identities)**.** *For every k,*

$$x \backslash c_k = c_{k-1}, \qquad c_k \backslash c_{k-1} = d_{k+1}.$$

If

$$p = x \backslash x = c_{d-1}, \qquad q = x \backslash p,$$

then

$$q = d_1 = c_{d-2} = (x \diamond x) \diamond x,$$

and

$$p \diamond (x \diamond x) = d_1, \qquad p \backslash d_1 = x \diamond x.$$

Proof. The first identity follows from $c_k = x \diamond c_{k-1}$. The second is precisely the recurrence in Lemma 5. Since $x \diamond c_{d-1} = c_d = x$, one has $p = c_{d-1}$, and then

$$q = x\backslash p = x\backslash c_{d-1} = c_{d-2} = d_1.$$

Also $d_1 = c_1 \diamond x = (x \diamond x) \diamond x$. Finally, Lemma 2, applied with $a = x$ and $b = x$, gives

$$x\backslash p = p \diamond (x \diamond x).$$

The left side is $q = d_1$, and hence $p \diamond (x \diamond x) = d_1$. The last displayed identity is the definition of left division by p. $\square$

Lemma 7 (A fixed point has the forced index)**.** *If $d_j = x$, then $j \equiv d - 2 \pmod{d}$.*

Proof. The condition $d_j = x$ says $c_j \diamond x = x$. Lemma 3 with $y = c_j, z = x$ gives

$$x = x \diamond (x \diamond c_j).$$

On the other hand, E677 with both variables specialized to x gives

$$x = x \diamond (x \diamond d_1).$$

Two left cancellations yield $c_j = d_1 = c_{d-2}$, so the minimality of the orbit period gives the claimed congruence. $\square$

Corollary 8 (No two-cycle in the L_x-orbit)**.** *The period d is never equal to* 2*. If $d = 1$, then* E255 at x *holds. Hence a counterexample point must have $d \geq 3$.*

Proof. If $d = 1$, then $x \diamond x = x$, and so

$$((x \diamond x) \diamond x) \diamond x = (x \diamond x) \diamond x = x \diamond x = x.$$

If $d = 2$, Lemma 5 gives $d_1 = c_{d-2} = c_0 = x$. But Lemma 7 applied with $j = 1$ would give $1 \equiv d - 2 \equiv 0 \pmod{2}$, a contradiction. $\square$

4 The route graph

The following display is the dependency graph used in the rest of the note. The first row contains equivalent fixed-point targets. The two lower branches describe the two ways an unfinished proof can still move: collision lifting or explicit counterexample construction.

$$\begin{array}{ccccccc}
\text{E255 at } x & \Longleftrightarrow & q \diamond x = x & \Longleftrightarrow & \exists u\ (u \diamond x = x) & \Longleftrightarrow & q\backslash x = x \\
 & & \Updownarrow & & & & \Updownarrow \\
 & & \text{unique witness } q & & & & ((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x \\
 & & & & & & \Updownarrow \\
 & & & & & & (q \diamond x) \diamond q = p.
\end{array}$$

If the fixed point is not established, the two honest remaining branches are

$$\begin{array}{ccccc}
d_i = d_j = e & \Longrightarrow & (c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j & \dashrightarrow & \text{finite counting pressure on } y \mapsto y \diamond e, \\
x \notin \operatorname{im} R_x & \Longrightarrow & y \diamond x = z \diamond x = r & \Longrightarrow & (y \diamond r) \diamond y = r\backslash x.
\end{array}$$

The dashed arrow is a missing lemma, not a theorem; the collision branch is now best viewed as a counting problem rather than a direct propagation problem.

5 The fixed-point frontier

The fixed-point route is sharper than the existential formulation suggests.

Proposition 9 (Unique right fixed-point witness)**.** *Let M be a finite* E677*-magma and fix $x \in M$. If $u \diamond x = x$, then*

$$u = q = (x \diamond x) \diamond x.$$

Consequently E255 *at x is equivalent to the existence of u with $u \diamond x = x$.*

Proof. The first half of the proof is the argument already used in Lemma 7. If $u \diamond x = x$, then Lemma 3 gives

$$x = x \diamond (x \diamond u).$$

The specialization $y = x$, $x = x$ in E677 gives

$$x = x \diamond (x \diamond q).$$

Two left cancellations give $u = q$. Since $q = (x \diamond x) \diamond x$, the identity $q \diamond x = x$ is exactly E255 at x. Thus E255 at x is equivalent to the existence of a right fixed point over x, and the witness is forced to be q. $\square$

Proposition 10 (Equivalent local targets)**.** *With the notation above, the following are equivalent:*

1. E255 *at x, equivalently $q \diamond x = x$;*
2. *$q \backslash x = x$;*
3. *$((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x$;*
4. *$(q \diamond x) \diamond q = p$.*

Proof. The equivalence of (1) and (2) is just uniqueness of left division by q. To identify the left quotient, apply Lemma 3 with $y = x \diamond x$ and $z = x$:

$$x = q \diamond \big(((x \diamond x) \diamond q) \diamond (x \diamond x)\big).$$

Therefore

$$q \backslash x = ((x \diamond x) \diamond q) \diamond (x \diamond x),$$

which proves the equivalence of (2) and (3), because $q = (x \diamond x) \diamond x$.

For (4), use Proposition 1:

$$q \backslash x = x \diamond ((q \diamond x) \diamond q).$$

Since $p = x \backslash x$, injectivity of L_x gives

$$x \diamond ((q \diamond x) \diamond q) = x \quad \Longleftrightarrow \quad x \diamond ((q \diamond x) \diamond q) = x \diamond p \quad \Longleftrightarrow \quad (q \diamond x) \diamond q = p.$$

Together with (1)$\Longleftrightarrow$(2), this proves the claim. $\square$

Definition 11 (Lemma L)**.** For the fixed element x, write $\mathrm{L}(x)$ for the identity

$$((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x.$$

By Proposition 10, $\mathrm{L}(x)$ is equivalent, in the present finite E677 setting, to E255 at x.

Remark 12. Lemma L is a target, not an established lemma. Because the unrestricted implication E677 $\models$ E255 is false [4, Section 13.2], any successful proof of the fixed-point route must use finiteness in a visible way: for example through left bijection, orbit periodicity, a finite-map lemma, or a collision argument. A derivation that simply rewrites E677 into Lemma L without using finite structure should be treated as suspect.

Lemma 13 (Fixed-map and edge-product form)**.** *For the fixed element x, define*

$$H_x(t) = ((x \diamond t) \diamond x), \qquad F_x(t) = x \diamond (t \diamond x).$$

Then

$$x \backslash t = t \diamond H_x(t) \tag{E}$$

for every $t \in M$, and

$$F_x(L_x(t)) = L_x(H_x(t)). \tag{Conj}$$

Consequently F_x and H_x have the same finite cycle structure. Moreover E255 *at x is equivalent to either of these maps having a fixed point. When fixed points exist, they are unique: the fixed point of F_x is*

$$x \backslash q,$$

and the fixed point of H_x is

$$x \backslash (x \backslash q).$$

Thus, at a counterexample point x, both maps are fixed-point-free and every cycle of either map has length at least 2.

Proof. Apply E677 with parameter x and law variable t:

$$t = x \diamond \big(t \diamond ((x \diamond t) \diamond x)\big) = x \diamond (t \diamond H_x(t)).$$

This is exactly (E). The conjugacy identity is immediate:

$$F_x(L_x(t)) = x \diamond ((x \diamond t) \diamond x) = L_x(H_x(t)).$$

Since L_x is bijective by Proposition 1, (Conj) gives the same cycle structure: L_x carries each H_x-cycle to an F_x-cycle, and L_x^{-1} carries each F_x-cycle back to an H_x-cycle.

If $F_x(z) = z$, then

$$x \diamond (z \diamond x) = z.$$

The left-inverse formula for L_x, applied to z, also gives

$$x \diamond \big(z \diamond ((x \diamond z) \diamond x)\big) = z.$$

Left cancellation by x, then by z, yields $(x \diamond z) \diamond x = x$. Hence a right fixed-point witness over x exists, and Proposition 9 gives E255 at x. More precisely, the same proposition gives

$$x \diamond z = q,$$

so any fixed point of F_x must be $z = x \backslash q$. Conversely, if $u \diamond x = x$, set

$$z = u \diamond ((x \diamond u) \diamond x).$$

Then $x \diamond z = u$ by E677 with parameter x, and another use of the same identity gives $z = x \diamond (z \diamond x)$, so $F_x(z) = z$. Taking the forced witness $u = q$, this fixed point is exactly $x \backslash q$. Finally, (Conj) shows that t is fixed by H_x if and only if $L_x(t)$ is fixed by F_x. Therefore the corresponding fixed point of H_x is $x \backslash (x \backslash q)$. $\square$

Remark 14. Identity (E) turns the fixed-point route into a finite dynamics problem. If $H_x(t) = t$, then $x\backslash t = t \diamond t$, equivalently $L_x(S(t)) = t$, which is the last fixed-point form in the ETP Lemma 13.2 list. In a bad point, the forward orbit of any element under H_x must eventually enter a cycle of length greater than 1. A finite proof via this route would have to show that such an H_x-cycle is incompatible with the edge products $x\backslash t = t \diamond H_x(t)$.

Remark 15 (Nontrivial H_x-cycles are not the obstruction). The previous remark should not be read as a global no-cycle target. In the standard linear model on $\mathbb{F}_5$,

$$u \diamond v = 2u - v,$$

one checks directly that E677 and E255 both hold. For fixed x,

$$H_x(t) = ((x \diamond t) \diamond x) = 3x - 2t.$$

Thus H_x fixes x, while on the other four elements it has a 4-cycle; for example, when $x = 0$, the cycle is

$$1 \mapsto 3 \mapsto 4 \mapsto 2 \mapsto 1.$$

So the finite-map target is not to rule out all nontrivial H_x-cycles. It is to rule out the fixed-point-free H_x-dynamics forced at a bad point, using the edge products $x\backslash t = t \diamond H_x(t)$ or an actual finite-map identity of the kind used in the ETP finite-map lemmas.

Remark 16 (Finite-map lemmas need a non-tautological input). The finite-map lemmas from the ETP finite chapter apply only after one has produced an identity of the form $f = f \circ f \circ g$ or $f = g \circ f \circ f$ for concrete self-maps of a finite set. The immediate identities

$$L_x = L_x \circ L_x \circ L_x^{-1}, \qquad L_x^{-1} = L_x^{-1} \circ L_x^{-1} \circ L_x$$

only recover the tautology $L_x = L_x \circ L_x^{-1} \circ L_x$. They do not involve the right translation R_x, the maps $H_x = R_x L_x$, $F_x = L_x R_x$, or the gate $t \mapsto (t \diamond x) \diamond t$. A future finite-map proof should therefore display a non-tautological identity for one of those maps before invoking Lemma 5.5 or Lemma 5.6.

6 Collision frontier

The orbit recurrence suggests a collision strategy, but the exact missing step is narrow.

Proposition 17 (Collision lift)**.** *If $d_i = d_j = e$, then*

$$(c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j. \tag{C}$$

If in addition $i \not\equiv j \pmod{d}$, then

$$c_i \diamond e \neq c_j \diamond e.$$

Proof. Lemma 3 with $y = c_k$ and $z = x$ gives

$$x = d_k \diamond ((c_k \diamond d_k) \diamond c_k).$$

If $d_i = d_j = e$, left cancellation by e yields (C). If also $c_i \diamond e = c_j \diamond e = s$, then (C) becomes $s \diamond c_i = s \diamond c_j$. Left cancellation by s gives $c_i = c_j$, contradicting the minimal period unless $i \equiv j \pmod{d}$. ∎

Problem 18 (Collision-counting target)**.** Turn a nontrivial collision $d_i = d_j = e$ into a finite counting contradiction, or into the fixed point $q \diamond x = x$. More precisely, let

$$F_e = \{y \in M : y \diamond x = e\}, \qquad A_e(y) = y \diamond e.$$

Proposition 17, and more generally Proposition 19, show that A_e is injective on F_e. Thus the old direct target $c_i \diamond e = c_j \diamond e$ is not a plausible standalone propagation lemma for a genuine collision: it is precisely what the splitter forbids unless a separate argument has already produced the fixed point. A collision proof should instead find a natural finite set B_e such that

$$A_e(F_e) \subseteq B_e$$

and $|B_e| < |F_e|$, unless $q \diamond x = x$.

The common shortcut "prove $k \mapsto d_k$ injective" is not by itself a proof of the main theorem. It becomes useful only if the argument also connects the resulting injectivity to the forced fixed point $q \diamond x = x$, or if it supplies the counting pressure requested in Problem 18.

7 Counterexample frontier

Suppose a finite counterexample exists and choose $x \in M$ with $q \diamond x \neq x$. Proposition 9 says that no element u has $u \diamond x = x$. Thus $x \notin \operatorname{im} R_x$. Since M is finite, R_x is not injective, and there exist $y \neq z$ and $r \neq x$ such that

$$y \diamond x = z \diamond x = r.$$

The following proposition records the forced shape of such a right collision.

Proposition 19 (Right-collision rectangles)**.** *Let $y \diamond x = r$. Then*

$$(y \diamond r) \diamond y = r \backslash x. \tag{R}$$

Consequently, if $y \diamond x = z \diamond x = r$ and $y \diamond r = z \diamond r$, then $y = z$.

Proof. Lemma 3 with $y = y$ and $z = x$ gives

$$x = r \diamond ((y \diamond r) \diamond y),$$

which is exactly (R). If $y \diamond r = z \diamond r = s$, then (R) gives $s \diamond y = s \diamond z$. Left cancellation by s gives $y = z$. □

This proposition is a counterexample design constraint. A candidate finite counterexample must contain right-collision fibers for every bad point x, but those fibers must split under the second probe $t \mapsto t \diamond r$, while all elements in the fiber preserve the same twisted diagonal value $(t \diamond r) \diamond t = r \backslash x$.

There are two plausible construction languages that respect this constraint. First, because all left translations are permutations, one may write

$$a \diamond b = \sigma(a)(b)$$

with $\sigma(a) \in \mathrm{Sym}(M)$. Law E677 is then the labeled- permutation condition

$$\sigma(y)\big(\sigma(x)(\sigma(\sigma(y)(x))(y))\big) = x.$$

A counterexample at x_0 would additionally require $\sigma(a)(x_0) \neq x_0$ for every label a. This is not merely a group-theoretic condition on a subgroup of $\mathrm{Sym}(M)$; the labeling map $a \mapsto \sigma(a)$ is part of the data. In fact this labeling map is injective. If $\sigma(y) = \sigma(z)$, choose any x and put $e = y \diamond x = z \diamond x$. The two specializations of E677 give

$$x = y \diamond (x \diamond (e \diamond y)), \qquad x = z \diamond (x \diamond (e \diamond z)).$$

Since $L_y = L_z$, the first equality may be rewritten with left factor z. Left cancellation by z, then by x, and finally by e, gives $y = z$. Thus a counterexample in this language is an injectively labeled set of permutations. In particular, if $y \diamond x = z \diamond x = r$ with $y \neq z$, the two distinct permutations $\sigma(y), \sigma(z)$ agree at x but must separate at r:

$$\sigma(y)(r) \neq \sigma(z)(r), \qquad \sigma(\sigma(y)(r))(y) = r \backslash x = \sigma(\sigma(z)(r))(z).$$

Second, one may seek nonlinear fiber extensions over positive E677-models. The ETP affine-extension obstruction [4, Lemma 13.4] rules out the obvious affine version, so a successful extension would have to be genuinely nonlinear and would still need to realize the rectangles in Proposition 19.

8 Obstructions and false routes

Several attractive shortcuts are invalid or unsupported.

1. The family
$$d_k \backslash x = c_{k-2} \diamond c_{k-1}$$
should not be used. The old derivation confused $c_{k-1} \diamond x$ with d_k; by definition it is d_{k-1}. The active repository does not contain a publication-ready finite table witness for this failure, so the present paper records only the derivation error and quarantines the identity.

2. The full-shift law
$$p \diamond d_k = d_{k+1}$$
is also quarantined. It is not a consequence of the trusted local package.

3. A proof of universal idempotence, $d = 1$, or any universal small-period conclusion would contradict known nontrivial positive models from the ETP finite theory. Such a claim should therefore be audited before it is used.

4. A proof that $H_x(t) = ((x \diamond t) \diamond x)$ has no nontrivial cycles is false as a global statement: Remark 15 gives a positive $\mathbb{F}_5$ model with nontrivial H_x-cycles. Only fixed-point-free bad-point dynamics is a live target.

5. A citation of the ETP finite-map lemmas is not by itself a finite proof. Those lemmas require a non-tautological self-map identity; the formal inverse identities for L_x and L_x^{-1} add no new information.

6. A nontrivial collision $d_i = d_j$ does not automatically propagate backwards to $d_{i-1} = d_{j-1}$. Any cascade proof must derive the propagation or avoid it.

9 Open frontiers

The work above leaves three precise frontiers.

Problem 20 (Fixed-point route)**.** Prove $q \diamond x = x$, equivalently Lemma L from Definition 11, by an argument that uses finite structure explicitly.

Problem 21 (Collision route)**.** Resolve Problem 18: from $d_i = d_j = e$, derive either the fixed point $q \diamond x = x$ or a finite counting contradiction for the injective splitter $A_e(y) = y \diamond e$ on the fiber $F_e = \{y : y \diamond x = e\}$.

Problem 22 (Counterexample route)**.** Construct a finite labeled-permutation model or a genuinely nonlinear fiber extension satisfying E677, violating E255, and satisfying the right-collision rectangle constraints of Proposition 19. A table-free parameter search is not a construction; any proposed example should include a complete finite table or a checkable structural verification.

A Proof audit checklist

The reductions in this note use left cancellation only after Proposition 1, where finiteness has made each L_a bijective. They do not use right cancellation, associativity, an identity element, or group/quasigroup structure. Future work on this problem should state which of the three frontiers it attacks and end, if unfinished, with one missing lemma rather than a broad search prescription.

Every new proof attempt should check the following list before being added to the mathematical record.

1. All products are parenthesized.
2. Every cancellation is a left cancellation by a known bijective left translation.
3. Every use of finiteness is named: left bijectivity, orbit periodicity, a finite-map lemma, or pigeonhole reasoning for a self-map.
4. The false quotient family and false full-shift law are not used.
5. Any counterexample claim contains either a complete finite operation or a structural verification of E677 and failure of E255.

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